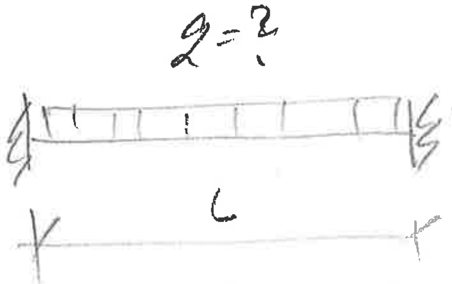
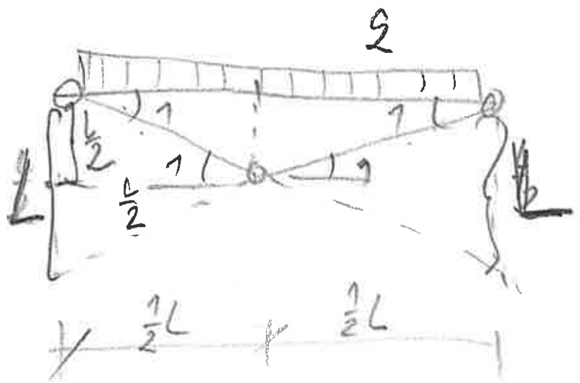
Upper Bound Theorem
Kinematic Theorem
UNSAVE



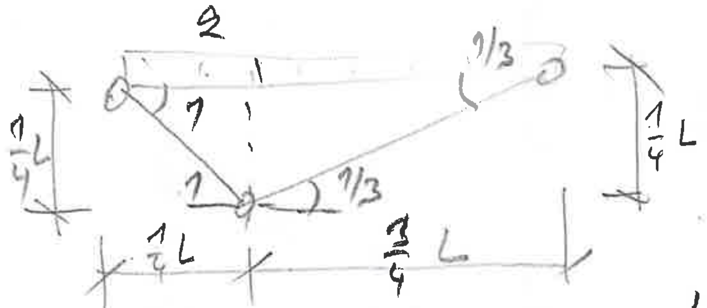


$W_{in} = M_{Rd}\cdot 4$

$W_{out} = 2\times q\cdot\frac{L}{2}\cdot\frac{1}{2}\left(\frac{L}{2}\right) = \frac{L^2}{4}q$

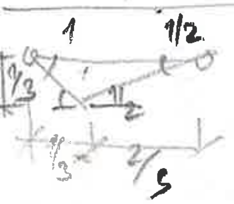
$W_{in} = W_{out} \Rightarrow M_{Rd}\cdot 4 = \frac{L^2}{4}q$

$$\boxed{q = \frac{16 M_{Rd}}{L^2}}$$



$W_{in} = M_{Rd}\left(2+\frac{2}{3}\right) = M_{Rd}\frac{8}{3}$

$W_{out} = q\cdot\frac{1}{4}L\cdot\frac{1}{2}\left(\frac{1}{4}L\right) + q\cdot\frac{3}{4}L\cdot\frac{1}{2}\left(\frac{1}{4}L\right) = qL^2\left(\frac{1}{32}+\frac{3}{32}\right) = \frac{4}{32}L^2 q$



$W_{in} = M_{Rd}(2+1) = 3M_{Rd}$

$W_{out} = q\frac{1}{3}L\cdot\frac{1}{2}\left(\frac{1}{3}L\right) + q\frac{2}{3}L\cdot\frac{1}{2}\left(\frac{1}{3}L\right) = qL^2\left(\frac{1}{18}+\frac{2}{18}\right) = \frac{1}{6}qL^2$

$W_{in} = W_{out} \Rightarrow 3M_{Rd} = \frac{1}{6}qL^2$

$$\boxed{q = 18\frac{M_{Rd}}{L^2}}$$

$W_{in} = W_{out} \Rightarrow \frac{8}{3}M_{Rd} = \frac{4}{32}L^2 q \Rightarrow$

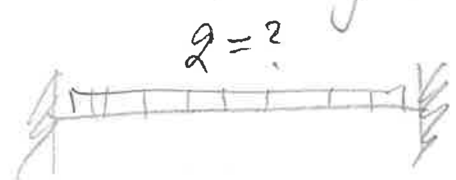
$\Rightarrow q = \frac{8}{3}\cdot\frac{32}{4}\frac{M_{Rd}}{L^2}$

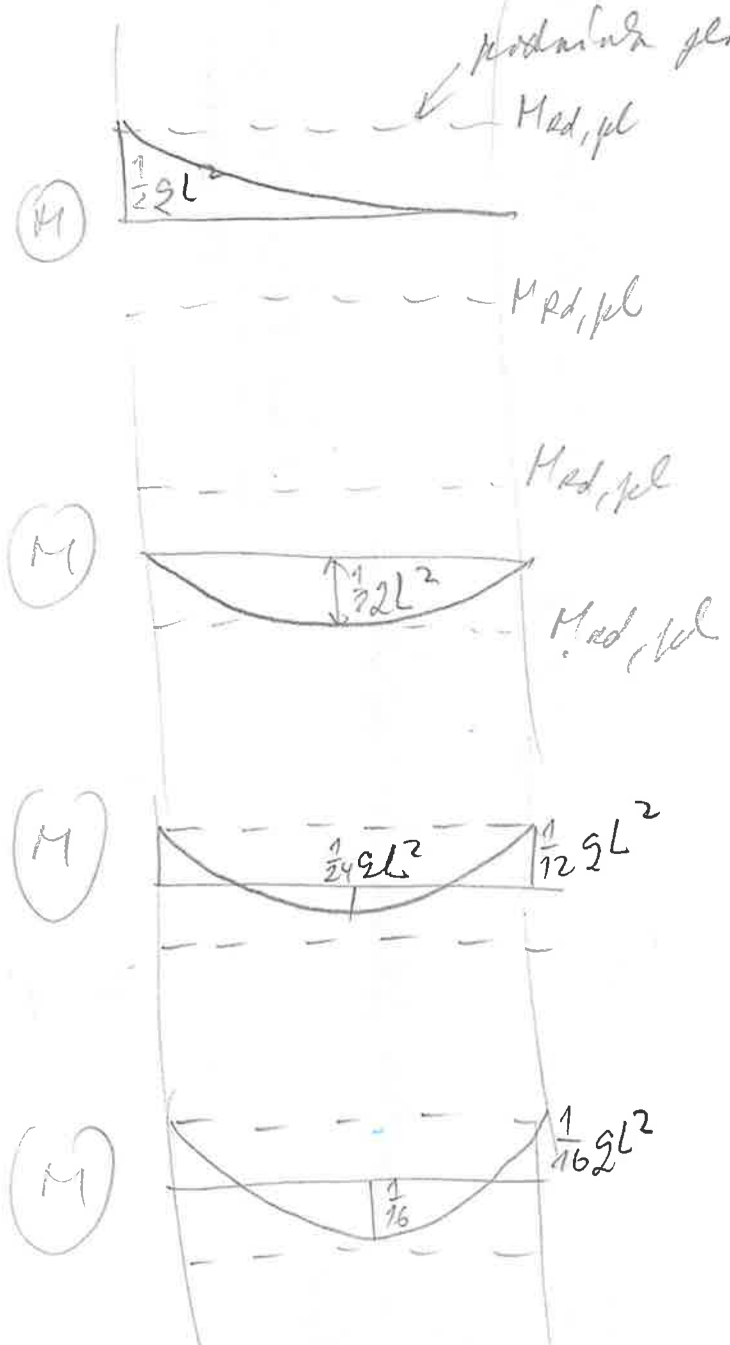
$$\boxed{q = \frac{64}{3}\frac{M_{Rd}}{L^2} = 21{,}\overline{3}\frac{M_{Rd}}{L^2}}$$

Z kinematicky přípustných stavů je třeba q minimalizovat.

Mezní analýza

Lower Bound Theorem
(Static Theorem)
SAVE





$M_{Rd} = \frac{1}{2}qL^2 \Rightarrow q = \frac{2M_{Rd}}{L^2}$

$M_{Rd} = \frac{1}{8}qL^2 \Rightarrow q = \frac{8M_{Rd}}{L^2}$

$M_{Rd} = \frac{1}{12}qL^2 \Rightarrow q = \frac{12M_{Rd}}{L^2}$

$M_{Rd} = \frac{1}{16}qL^2 \Rightarrow q = \frac{16M_{Rd}}{L^2}$

Z přípustných stavů, kdy jsou splněny podmínky rovnováhy (Schwedlerovy věty) a podmínky plasticity ($M_{Ed} \le M_{Rd,pl}$) ~~tak~~ je ~~max~~ úkolem konstrukce ~~stav maximalizace~~ možno maximalizovat zatížení pro tyto přípustné stavy.